

Deep Reinforcement Learning Cheat Sheet

BITS Pilani WILP

Prepared with ChatGPT Assistance

June 28, 2025

1. MDP Framework

- **Components:** States (S), Actions (A), Rewards (R), Transition Probabilities (P), Discount Factor (γ).
- **Bellman Expectation (Value Function):**

$$V(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) [R + \gamma V(s')]$$

- **Bellman Optimality (Value Iteration):**

$$V(s) = \max_a \sum_{s'} P(s'|s, a) [R + \gamma V(s')]$$

- **Action-Value Function:**

$$Q(s, a) = \sum_{s'} P(s'|s, a) [R + \gamma V(s')]$$

2. Dynamic Programming

- **Policy Evaluation:**

$$V(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) [R + \gamma V(s')]$$

- **Policy Improvement:**

$$\pi'(s) = \arg \max_a \sum_{s'} P(s'|s, a) [R + \gamma V(s')]$$

- **Value Iteration:** Iteratively apply Bellman Optimality until convergence.
- **Policy Iteration:** Alternate between Policy Evaluation and Policy Improvement until stable.

3. Bandit Algorithms

- **Sample Average Update:**

$$Q_{n+1} = Q_n + \frac{1}{N} (R - Q_n)$$

- **Weighted Average Update (Non-stationary):**

$$Q_{n+1} = Q_n + \alpha (R - Q_n)$$

- **ϵ -Greedy:** Exploit best Q with $(1 - \epsilon)$ probability, explore randomly with ϵ probability.
- **UCB (Upper Confidence Bound):**

$$Q + c\sqrt{\frac{\ln t}{N(a)}}$$

- **Optimistic Initialization:** Start with high Q-values to encourage exploration.

4. Monte Carlo Methods

- **First-Visit MC:** Update $V(s)$ only on first visit per episode.
- **Every-Visit MC:** Update $V(s)$ every time state s is visited.
- **Return G:**

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

- **Importance Sampling (IS):**

$$\rho = \prod \frac{\pi(a|s)}{b(a|s)}$$

- **Exploring Starts:** Start from random (state, action) pairs to ensure exploration.

5. Numerical Problem Patterns

- **ϵ -Greedy:** If selected action is not the max Q, ϵ definitely occurred.
- **Q Updates:**

$$Q = \frac{\text{Initial Q} + \text{Sum of Rewards}}{\text{Count} + 1}$$

- **Value Iteration Formula:**

$$V(s) = \max_a \sum_{s'} [P(s'|s, a)(R + \gamma V(s'))]$$

- **Return Calculation:** Sum rewards with discount factor γ .

6. Formula Sheet

- Bellman Value:

$$V(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) [R + \gamma V(s')]$$

- Bellman Optimality:

$$V(s) = \max_a \sum_{s'} P(s'|s, a) [R + \gamma V(s')]$$

- Action-Value:

$$Q(s, a) = \sum_{s'} P(s'|s, a) [R + \gamma V(s')]$$

- Return:

$$G = R_1 + \gamma R_2 + \gamma^2 R_3 + \dots$$

- Sample Average:

$$Q_{n+1} = Q_n + \frac{1}{N} (R - Q_n)$$

- Weighted Average:

$$Q_{n+1} = Q_n + \alpha (R - Q_n)$$

- Importance Sampling:

$$\rho = \prod \frac{\pi(a|s)}{b(a|s)}$$